

PAPER CODE: BET-C620
EVEN SEMESTER EXAMINATION (MAY-2026)
CLASS: B.TECH. SEMESTER: VI
PAPER TITLE: EMBEDDED SYSTEM AND IOT APPLICATIONS

Time: 3 hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
 (5 x 6 = 30 Marks)

		CO	BL
Q-1 -	Define an Embedded System. Explain any four applications of embedded systems.	CO1	BL2
Q-2 -	Explain the working and importance of a watchdog timer in embedded systems.	CO1	BL2
Q-3 -	Describe the concept of hardware and software partitioning in embedded system design.	CO1	BL3
Q-4	Compare AVR and PIC microcontrollers with suitable examples.	CO2	BL3
Q-5	Explain the important features of ESP32 and ESP8266 microcontrollers.	CO2	BL2
Q-6 -	Compare CISC and RISC processor architectures with suitable examples.	CO3	BL4
Q-7	Discuss the addressing modes of microcontroller 8051 with an example.	CO3	BL5
Q-8 -	Explain Direct Memory Access (DMA) and its advantages in data transfer.	CO4	BL2
Q-9	Draw the interfacing circuit and write a program in C language to interface Stepper motor with microcontroller 8051.	CO4	BL3
Q-10	Describe different IoT communication technologies such as Wi-Fi, BLE and Zigbee.	CO5	BL2

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 x 10 = 40 Marks)

		CO	BL
Q-1	Explain the embedded system design life cycle with a neat diagram.	CO1	BL3
Q-2	Write short notes on following: (a) I/O sinking and Sourcing (b) Push Buttons interfacing with 8051 controller.	CO1	BL1
Q-3	Explain the I/O ports and memory organization of the 8051 microcontroller with a diagram.	CO3	BL2
Q-4	Describe the SPI and I ² C serial communication protocols used in embedded systems.	CO4	BL3
Q-5	Describe timers/counters and serial communication in 8051.	CO3	BL3
Q-6	Draw the interfacing circuit and write a program in C language to interface Common Cathode Seven segment LED with microcontroller 8051.	CO3	BL4
Q-7	Explain IoT architecture and IoT communication models such as device-to-device, device-to-cloud and gateway-based communication.	CO5	BL3
Q-8	Discuss LoRaWAN and NB-IoT technologies and their applications in IoT.	CO5	BL3
